Olist Brazilian E-Commerce: A Comprehensive Data Analysis

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Introduction and Objectives

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- The dataset used in this analysis is the Brazilian E-Commerce Public Dataset provided by Olist, the largest department store in Brazilian marketplaces. It contains information on 100,000+ orders placed between 2016 and 2018, covering customer behavior, sales performance, and logistics across Brazil.

- This analysis focuses on three key objectives:
  - 1. Understand Customer Behavior
  - 2. Evaluate Sales Performance
  - 3. Examine Logistical Efficiency

Methodology

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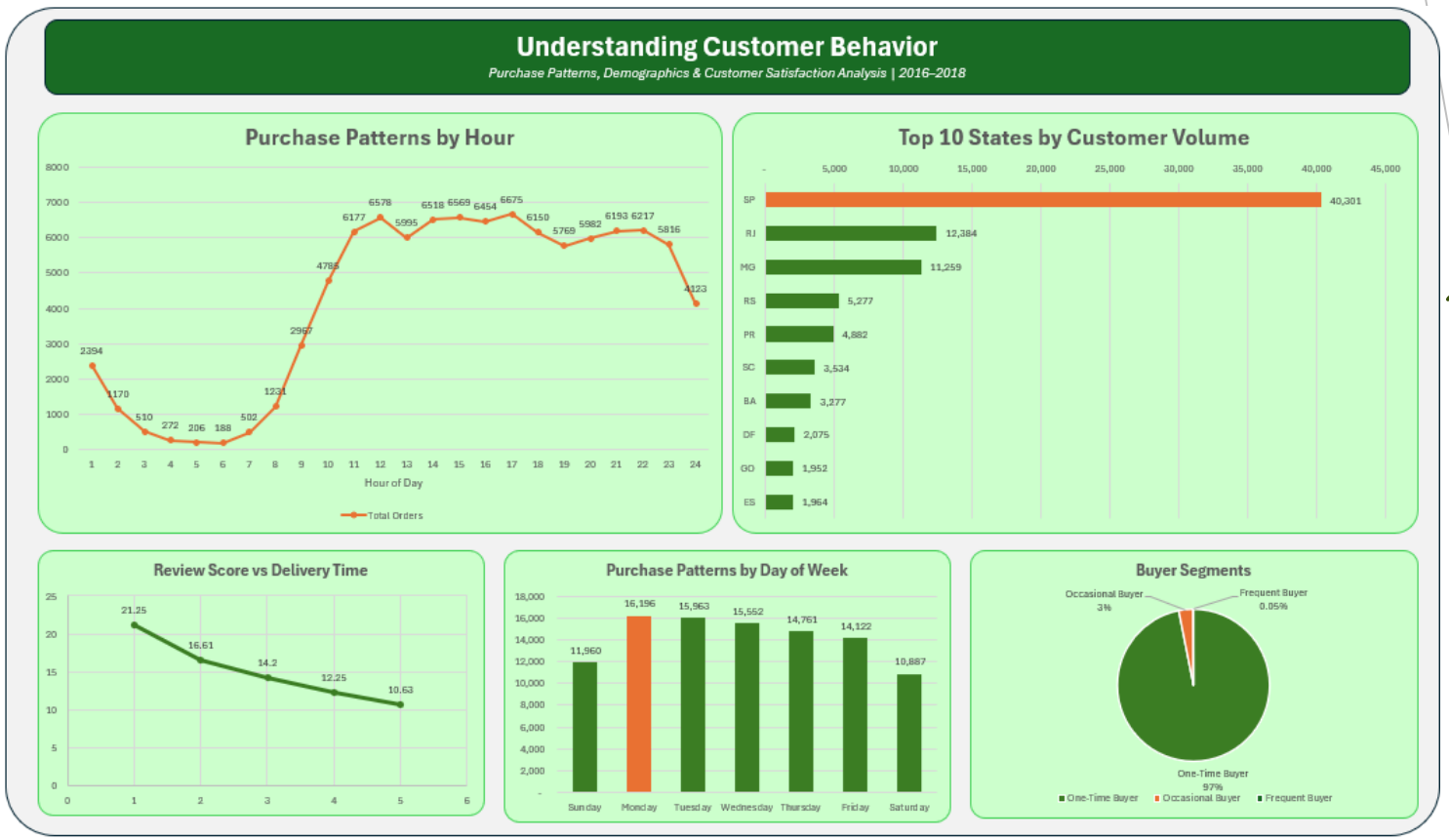
- **Tools Used**
  - MySQL 8.0 Workbench (Data extraction and analysis)
  - Microsoft Excel (Data visualization and charting)
- **Data Preprocessing Steps**
  - Imported 11 CSV datasets into MySQL database
  - Handled NULL values and blank cells using NULLIF()
  - Fixed date formats using STR_TO_DATE()
  - Cleaned text columns using TRIM()
  - Resolved Windows line ending issues (\r\n)
  - Joined tables based on ERD relationships
- **Analysis Approach**
  - Wrote SQL queries for each objective
  - Exported results to Excel
  - Created visualizations for each finding
  - Created insights and recommendations

Results of Analysis

Results (Objective 1: Understand Customer Behavior)

Orders peak between **10AM to 5PM** with 6,000+ orders per hour, and drop to their lowest between **3AM to 5AM.**



São Paulo dominates with 40,301 customers which is 3x more than Rio de Janeiro. The southeastern region drives the majority of Olist's customer base.

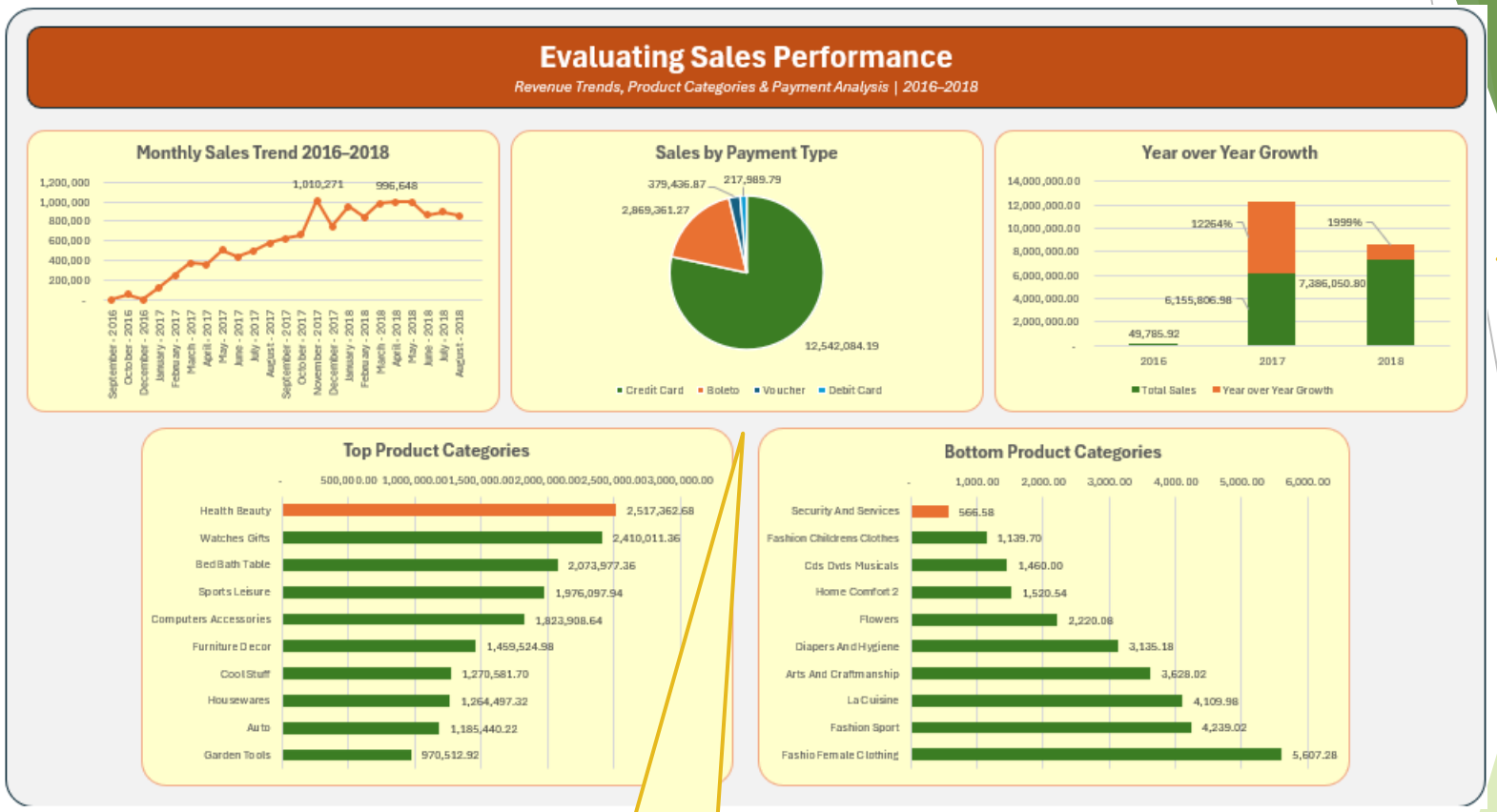
Faster delivery = higher satisfaction. Orders delivered in **~10 days earn 5 stars**, while those taking **~21 days earn only 1 star.**

Monday is the busiest day with 16,196 orders. Weekends, especially Saturday, record the lowest activity at 10,887 orders.

97% of customers only buy once. With only 0.05% frequent buyers, customer retention is Olist's biggest challenge.

Results (Objective 2: Evaluating Sales Performance)

Sales show **consistent upward growth** from 2016 to 2018. **November 2017 peaked at R$1,010,271**, likely driven by Black Friday. Growth remained strong in 2018 averaging R$900K+ per month.



Revenue grew from **R$49,785 in 2016 to R$7,386,050 in 2018**. The more reliable **2017 to 2018 growth is +19.99%**, showing sustainable business expansion. 2016 reflects only 3 months of data from the dataset.

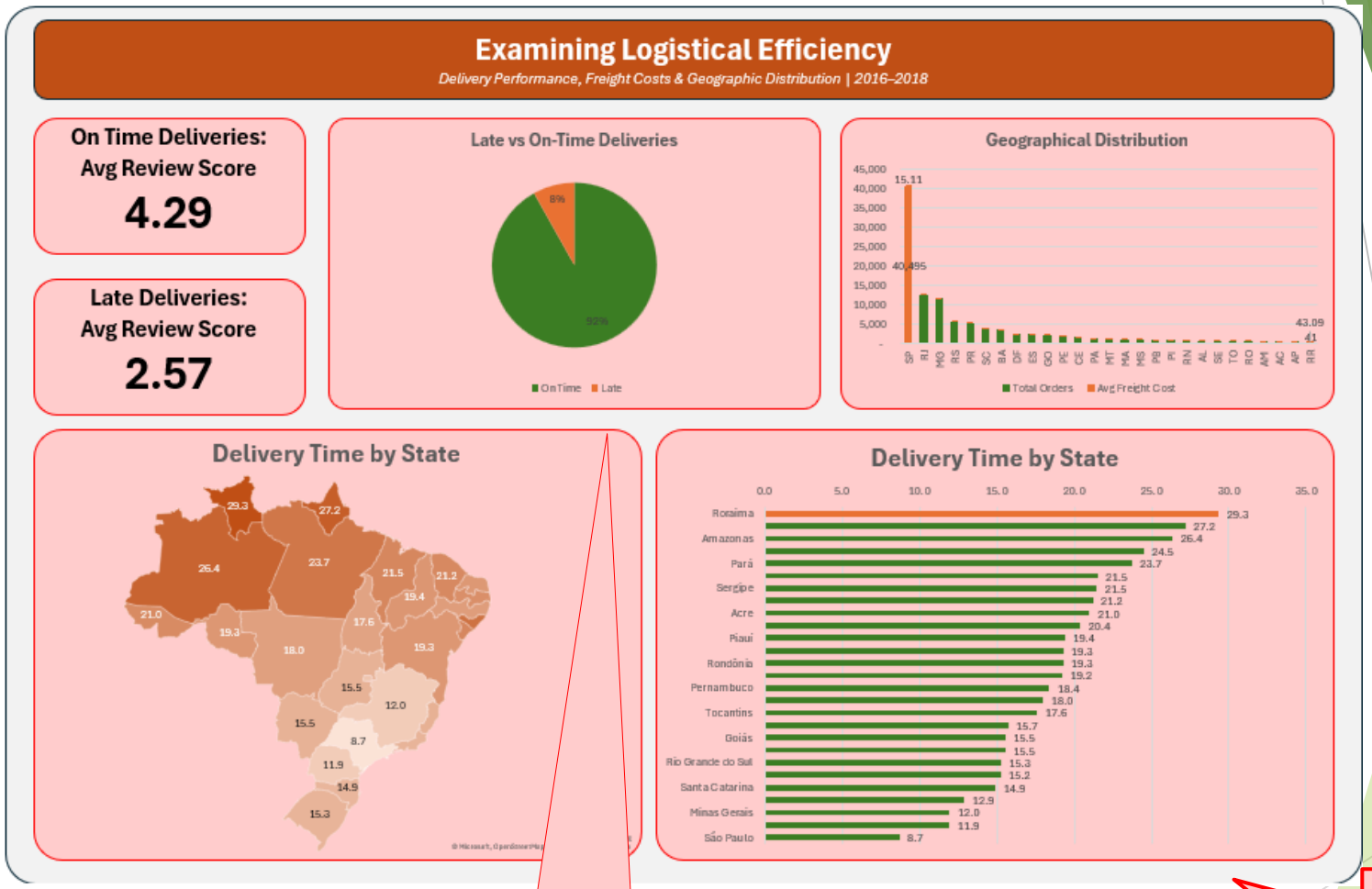
Health Beauty leads with R$2,517,362 in revenue, followed by Watches & Gifts (R$2,410,011) which makes the **highest average price at R$201**. These two categories alone drive significant revenue.

Credit Card have the highest revenue with R$12,542,084 — representing approximately **78% of total revenue**. Boleto is a distant second at R$2,869,361, while Voucher and Debit Card contribute minimally.

Security & Services has the lowest revenue at just R$566 from only 2 orders. The bottom categories show extremely low demand.

Results (Objective 3: Examining Logistical Efficiency)

On-time deliveries gets an average score of **4.29** while late deliveries drop to **2.57** which is a **40% decrease in satisfaction.** Delivery speed is the single most critical factor affecting customer experience.



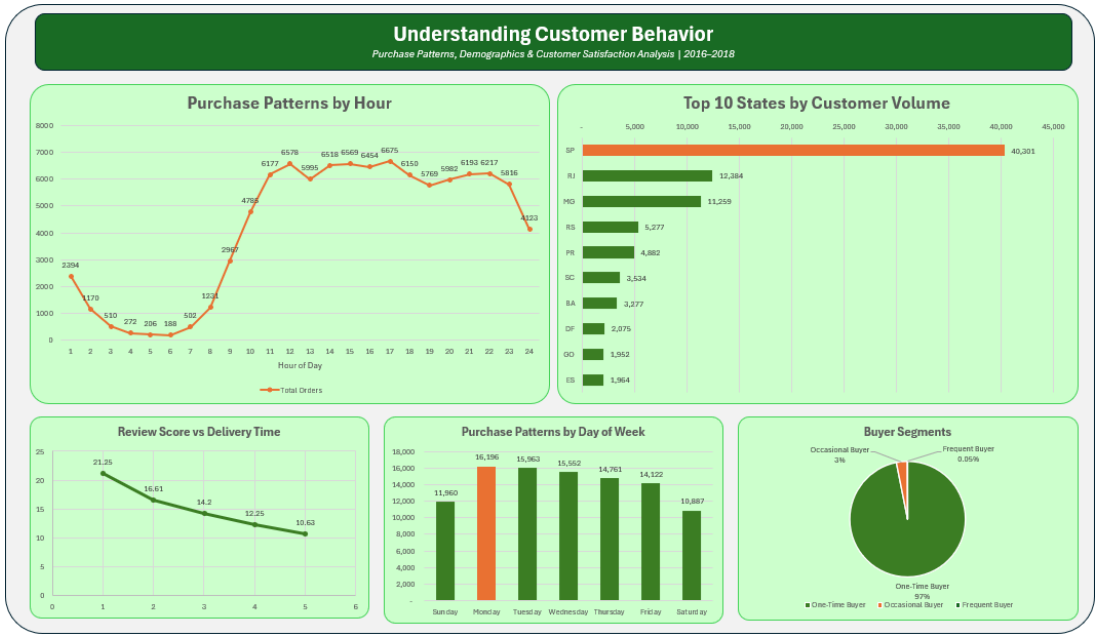
São Paulo leads with 40,495 orders and the **lowest average freight cost at R$15.11.** Remote states like Roraima pay nearly **3x more in freight (R$43.09)**, creating a significant cost barrier for customers outside major cities.

92% of orders are delivered on time, with only **8% arriving late.** While the on-time rate is strong, the 8% late rate still represents thousands of unsatisfied customers directly impacting review scores.

A clear **north-south difference** exists in delivery performance. **São Paulo delivers fastest at 8.7 days** while **Roraima takes 29.3 days** — more than 3x longer. Northern and remote states consistently experience the slowest deliveries due to limited logistics infrastructure.

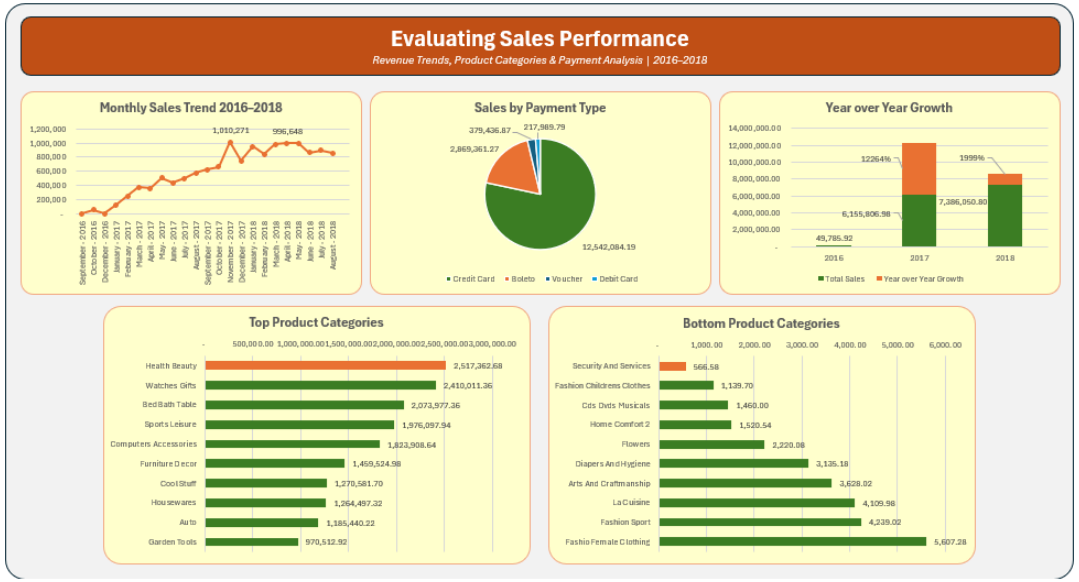
Recommendations

Recommendation (Objective 1: Understand Customer Behavior)



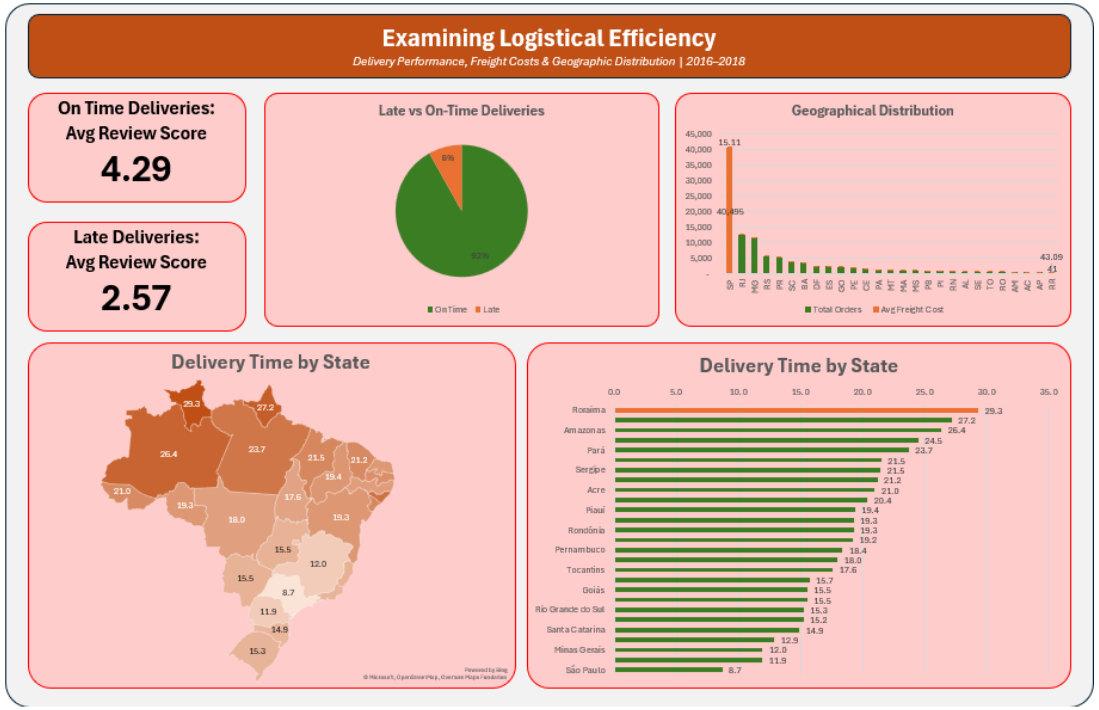
- **Launch a Loyalty Program**
  - Target the 96.88% one-time buyers with rewards and personalized offers to get more repeat purchases.
- **Optimize Marketing Timing**
  - Schedule promotions on **Monday to Tuesday between 10AM to 4PM** to maximize sales during peak hours and days.
- **Target High-Value Regions**
  - Focus premium campaigns on high average order value states like **Bahia** alongside São Paulo.

Recommendation (Objective 2: Evaluating Sales Performance)



- **Invest in Top Categories**
  - Put more marketing weight and inventory behind the two strongest revenue drivers: *Health*, *Beauty*, and *Watches & Gifts*.
- **Review Low-Performing Categories**
  - Assess categories with very low sales, such as Security & Services, and consider improving or phasing them out.
- **Leverage Credit Card Dominance**
  - Work with payment partners to offer installment options or cashback to encourage higher spending.

Recommendation (Objective 3: Examining Logistical Efficiency)



- **Expand Northern Logistics Coverage**
  - Set up distribution centers or local courier partnerships in **Roraima and Amazonas** to reduce the 29+ day delivery times.
- **Implement Delivery Monitoring**
  - Build an alert system to identify late deliveries early and notify customers before complaints appear.
- **Reduce Remote Freight Costs**
  - Negotiate bulk shipping rates and explore freight allowances for states paying **R$40+ in average freight costs.**

Conclusion

- Olist has demonstrated **strong and consistent growth** from 2016 to 2018, with revenue increasing **19.99% year over year**. The analysis uncovered three critical priorities which are:
  - **Retention** - 96.88% one-time buyer rate must be addressed through loyalty initiatives
  - **Category Improvement** - Focus more on top performers, reassess bottom categories
  - **Logistics Performance** - Close the delivery gap between São Paulo (8.7 days) and northern states (29+ days)
- If Olist addressed these issues, they can then:
  - Convert one-time buyers into repeat customers.
  - Get higher revenue by focusing resources on high-performing categories and peak buying hours.
  - Reduce operational costs through better shipping partnerships and negotiations.
  - Improve customer satisfaction by shortening delivery windows in underserved regions.